June 19, 1959

MEMO TO: Dr. D. V. Newton

SUBJECT: Assignment of Product Names for the 7030 (Stretch) System

I have prepared the following breakdown of the elements of the 7030 (Stretch) System together with suggested names for the various units. These are the smallest functional units into which the system can be divided.

Suggested Name	Description
Card Reader	This is a 1000 card per minute Card Reader which is part of the I/O System
Card Punch	This is a 250 card per minute Card Punch which is part of the I/O System
Printer	This is a 600 line per minute Chain Printer which is part of the I/O System
Card Reader Control Unit	This unit contains the buffers and electronics which control the Card Reader and make it attachable to an Exchange channel.
Card Punch Control Unit	This unit contains the buffers and electronics which control the Card Punch and make it attachable to an Exchange channel
Printer Control Unit	This unit contains the buffers and electronics which control the printer and make it attachable to an Exchange channel
Console	This unit contains the elements of the console for operator control of the system including a modified 625 typewriter

Suggested Name	Description
Console Control Unit	This unit contains the buffers and electronics for control of the console which make it attachable to an Exchange channel
Core Storage	This unit is identical to the Core Storage Unit announced with the 7090 Data Processing System. 1,2,4,6,8,10,12, or 16 units can be attached to the system
Central Processing Unit	The Central Processing Unit performs arithmetic and logical operations for operands taken from storage. Operations are specified one at a time by instructions which are also taken from storage
Exchange	The function of the Exchange is to direct the information flow between input-output or external storage units and internal storage. The Exchange enables processing of data to proceed simultaneously with the operation of a number of input-output units, being able to transmit data to and receive data from any part of the main internal storage. The Exchange has a minimum of 8 channels. Additional channels can be added in blocks of eight up to a total of 32 channels.
Low Speed Exchange	The Low Speed Exchange is a further extension of the Exchange which allows a great many more units of much lower speed to be operated simultaneously. A Low Speed Exchange has a minimum of 16 channels. Each of which can receive data at a maximum rate of 250 characters per second. An additional group of 16 channels can be added. Up to two Low Speed Exchanges can be added to the Exchange.
High Speed Exchange	The purpose of the High Speed Exchange is to connect disk storage units to the system. The High Speed Exchange has only one channel for data transmission. Consequently, only one unit can be involved in data transmission at a time. It can accommodate, however, a total of 32 units.

Suggested Name	Description
Disk Storage Unit	This unit contains 4,194,304 words of storage. Data can be written onto or read from this unit at the rate of 250,000 words per second
Tape Control Unit	This unit controls up to 8 magnetic tape units and contains the buffers and electronics which make them attachable to an Exchange channel. There will be two models. One for attaching dual density 729-II tape units and another for attaching dual density 729-IV units
Power Distribution Unit	This unit is the main power distribution point for 60 cycle and 400 cycle power to the 7030 system. It also contains circuitry to aid maintenance of the core storage units attached to the system

A 7030 Data Processing System must contain a Central Processing Unit, an Exchange, a Power Distribution Unit, one or more Core Storage Units, a Card Reader and a Printer. A minimum operational system will, of course, contain many other units including several Magnetic Tape Units, at least one Console, and probably a Disk Storage Unit together with their Control Units.

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